

Project Idea Summary

Project Title:

Automated Phishing Email Detection System Using Machine Learning

Project Idea:

The idea of this project is to build an intelligent system that can automatically detect phishing emails using machine learning techniques. Phishing emails are fraudulent messages designed to trick users into revealing sensitive information such as passwords or financial details. Traditional filters often fail to detect modern phishing attacks, which rely on social engineering and carefully written messages.

This project uses data-driven approaches and Natural Language Processing (NLP) to analyze email content and classify emails as either *phishing* or *legitimate*.

Motivation:

Phishing attacks are one of the most common cybersecurity threats today. They affect individuals, organizations, and institutions, causing financial loss and data breaches. There is a strong need for automated and intelligent solutions that can accurately detect phishing emails without relying on manual rules or external paid services.

Proposed Solution:

The proposed system processes email text and extracts important features using NLP techniques such as TF-IDF. These features are then passed to machine learning models trained on real phishing datasets. The system outputs a clear decision indicating whether the email is phishing or safe, along with a confidence score.

The solution is lightweight, privacy-preserving, and can work offline.

Tools & Technologies:

- Python

- Machine Learning (scikit-learn)
- Natural Language Processing (NLP)
- Public phishing email datasets

Expected Outcomes:

- A trained machine learning model capable of detecting phishing emails.
- Improved accuracy compared to traditional rule-based filters.
- A practical application of data science and cybersecurity concepts.

Project Scope:

The project focuses on email text analysis and supervised machine learning classification. It does not depend on external APIs or browser extensions, making it suitable for academic and practical use.

Conclusion:

This project demonstrates how machine learning and NLP can be effectively applied to solve real-world cybersecurity problems. The final system provides an efficient and scalable approach to phishing email detection.